



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	AAI

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	6
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	03
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	07

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Vaishali Shirsat

Signature :

Date :

6/4/25

Q. 1) If a retail company wants customers to visualise how furniture would look inside their real home using their smartphone camera. Identify which technology will be helpful and explain how it works.

→ Technology: Augmented Reality (AR) -

- Augmented reality (AR) overlays virtual objects onto the real-world environment using a smartphone camera or AR-enabled device. In this case, customers can see how furniture would look in their actual home space.

- How it works?

- i) The smartphone camera captures the real-world environment.
- ii) AR software detects surfaces (floor, walls, space) using computer vision and sensors.
- iii) A 3D model of furniture is placed in the detected space.
- iv) The object is scaled, rotated, and adjusted in real time.
- v) The user can interact (move, resize, change color).

- Key benefits -

- Realistic visualization before purchase.
- Improves customer confidence.
- Reduces product returns.

- The smartphone camera captures the real-world environment.

- AR sw detects surface using computer vision and sensors.

- A 3D model of furniture is placed in the detected space.

- The object is scaled, rotated and adjusted in real time.

- The user can interact.

- AR sw detects camera captures the real-world environment.

Q. 2) If a company wants to create a digital platform where users interact through avatars, attend meetings, buy virtual assets, participate in an events continuously. Identify which technology he helpful and explain how it works?

→ Technology: Metaverse

Exple:-

The metaverse is a persistent virtual world where users interact through avatars, attend meetings, socialize and trade digital assets.

How it works -

- i) Users create avatars representing themselves.
- ii) They enter a shared virtual environment using VR/AR or web platform.
- iii) The system runs continuously.
- iv) Users can:
 - Attend meetings / events
 - Buy / Sell virtual assets (NFTs, digital goods)
 - Interact socially in real time.
- v) Powered by technologies like:
 - Virtual Reality
 - Blockchain (for ownership)
 - AI (for personalization)

Key Benefits -

Immersive collaboration

Digital economy

Continuous engagement

Metaverse building

Trade digital assets.

Q. 3) If a scientist wants to solve extremely complex optimization problems in AI much faster than classical computers allow, identify which emerging technology can be used and explain how it helps.

→ Technology - Quantum computing -

- Explanation -

- Quantum computing uses principles of quantum mechanics to process information much faster than classical computers for certain problems.

- How it helps -

- Uses qubits instead of classical bits.

- Leverages - Superposition - multiple state at once.

- entanglement - correlation qubits.

- can explore many possible solutions simultaneously.

- Applications in AI optimization -

- faster training of complex models.

- Solving combinatorial optimization problems.

- Improved search and sampling.

- why it's powerful -

- Exponential speedup for certain problems.

- Handles high-dimensional data efficiently.

- faster training of complex models.

- solving combination optimization problem.

- Improved search and sampling.

- Exponential speedup for certain problems.

- Handles high-dimensional data efficiently.

Q. 4) what is metaverse? Explain the characteristics and components of the metaverse.

→ The metaverse is a shared, persistent, immersive universe where users interact with each other and digital environments in real time using avatars.

characteristics of metaverse -

(i) Persistence -

The virtual world continues to exist even when users are offline.

(ii) Real-time interaction -

Users interact instantly with others and the environment.

(iii) Immersiveness -

Provides a highly engaging experience using VR/AR.

(iv) Interoperability -

Asset and identities can move across platforms.

Components of metaverse -

(i) Virtual Reality (VR) -

fully immersive digital environment.

(ii) Augmented Reality (AR) -

Blends real and virtual worlds.

(iii) Blockchain -

ensures ownership of digital assets (NFTs, cryptocurrencies).

(iv) Artificial Intelligence - enables smart interaction, avatars, personalize.